

Name	Rollno	Section
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ASSIGNMENT# 01

Question#1:

Define the process involved in the analysis of Bubble sort algorithm through derivation, code algorithm in C++ or C# or Python.

Definition:

Bubble Sort is a **comparison-based sorting algorithm**. It works by **repeatedly swapping adjacent elements if they are in the wrong order**, bubbling the largest element to the end in each pass.

Analysis:

Bubble sort Algorithm	Time Complexity
int Bubble sor(int arr[])	
{	
int n=arr.length;	1
int temp ;	1
for (i=0 ; i < n-1 ; i++)	$1+N-1+1+N-1 = 2N$
{	
for (j=0 ; j < n-1 ; j++)	$1+x+1+x = 2x+2$
{	
if(arr[j] > arr[j+1])	$3x$
{	
temp = arr[j];	$2x$
arr[j]=arr[j+1];	$2x$
arr[j+1] = temp;	$2x$
}	
}	
}	
return arr;	1
}	
	$T(N) = 1+1+2N+2x+2+3x+2x+2x+2x+2$ $T(N) = 11x + 2N + 6$... $X = 1 + 2+3 + \dots (N-1)$ in Arithmetic progression
	Considering AP $1+2+3+\dots k = K(K+1)/2$ $X = (n-1)(n-1+1)/2$

	$T(N) = 11/2 N^2 - 11/2N + 2N + 6$
	Considering big O notation and ignoring constant we consider the term of higher growth.
	$T(N) = O(N^2)$.

Question#2:

Define the process involved in the analysis of Selection sort algorithm through derivation, code algorithm in C++ or C# or Python.

Definition:

Selection Sort is a simple sorting algorithm that divides the array into a sorted and an unsorted part. Initially, the sorted part is empty. The algorithm repeatedly selects the smallest element from the unsorted part and moves it to the sorted part. This process continues until the entire array is sorted. Its time complexity is $O(n^2)$.

Analysis:

Selection sort Algorithm	Time Complexity
int Selection sort (int arr []) {	
int minindex;	1
int temp;	1
for (int i=0; i < n-1;i++)	$1+N-1+1+N-1 = 2N$
minindex= i;	$N-1$
for(int j = i+1; j < n ; j++) {	$N-1+x+1+x$
if (a [j] < a [minindex]) {	$3x$
minindex = j; } } }	x
temp = arr[i]	$2(N-1)$
arr[i] = arr[minindex]	$2(N-1)$
arr[minindex] = temp; }	$2(N-1)$
return arr; }	1
	$T(N) = 1+1+2N+N-1+N-1+x+1+x+3x+x+2(N-1)+ 2(N-1)+ 2(N-1)+1 = 6x+10N-4$
	... $X = 1 + 2+3 + \dots (N-1)$ in Arithmetic progression Considering AP $1+2+3+\dots k = K(K+1)/2$

	$X = (N-1)(N-1+1)/2$ $6N^2/2 - 6N/2 + 10N - 4$ $T(N) = 3N^2 - 3N + 10N - 4$ $T(N) = 3N^2 + 7N -$
	Considering big O notation and ignoring constant we consider the term of higher growth.
	$T(N) = O(N^2)$.

Question#3:

Define the process involved in the analysis of Insertion sort algorithm through derivation, code algorithm in C++ or C# or Python.

Definition:

Insertion sort is a simple sorting algorithm that builds a sorted list one element at a time by comparing and inserting each element into its correct position. It's efficient for small or nearly sorted datasets, with a best-case time complexity of $O(n)$ and worst-case of $O(n^2)$.

Analysis:

Insertion Sort Algorithm	Time Complexity
Int insertion sort (int a[])	
{	
int i;	1
int i;	1
int temp;	1
}	
for(i=0; i < n; i++)	$1+N-1+1+N-1 = 2N$
}	
temp = a[i]	$2(N-1)$
j = i-1;	$2(N-1)$
While(j >= 0 && a[j] > temp)	$3x$ (assume $N-1=x$)
{	
A[j+1] = a[j];	$2x$
j--;	x
}	
A[j+1] = temp;	$2(N-1)$
}	
}	
	$T(N) = 1+1+1+2N+2N-2+2N-2+3x+2x+x+2N-2$ $6x+8N+1$

	... $X = 1 + 2 + 3 + \dots + (N-1)$ in Arithmetic progression
	Considering AP $1 + 2 + 3 + \dots + k = K(K+1)/2$
	$= 6N^2/2 - 6N/2 + 8N + 1$
	$= 3N^2 - 3N + 8N + 1$
	Considering big O notation and ignoring constant we consider the term of higher growth.
	$T(N) = O(N^2)$.

Question#4:

Define the process involved in the analysis of Shell sort algorithm through derivation, code algorithm in C++ or C# or Python.

Definition:

Shell sort is an improved version of insertion sort that sorts elements at specific intervals (gaps), gradually reducing the gap until the final pass. It allows faster movement of elements toward their correct positions and performs better than basic sorting algorithms for medium-sized datasets.

Analysis:

Shell Sort Algorithm	Time Complexity
int shellSort(int arr[], int n) {	
for (int gap = n/2; gap > 0; gap /= 2) {	...?
for (int i = gap; i < n; i += 1) {	$1 + N - 1 + 1 + N - 1 = 2N$
int temp = arr[i];	2x
int j;	1
for (j = i; j >= gap && arr[j - gap] > temp; j - = gap) {	?
arr[j] = arr[j - gap]; }	2x
arr[j] = temp; } } }	2x

Question#5:

Define the process involved in the analysis of Heap sort algorithm through derivation, code algorithm in C++ or C# or Python.

Definition:

Heap sort is a sorting algorithm that builds a binary heap from the data, then repeatedly removes the largest element from the heap and places it at the end of the array. It maintains the heap property after each removal until the entire array is sorted. It runs in $O(n \log n)$ time and sorts in place without extra memory.

Analysis:

Heap Sort Algorithm	Time Complexity
heapify(int arr[], int n, int i)	
{	
int largest = i;	2
int left = 2 * i + 1;	2
int right = 2 * i + 2;	2
if (left < n && arr[left] > arr[largest])	
largest = left;	x
if (right < n && arr[right] > arr[largest])	
largest = right;	x
if (largest != i)	
{	
swap(arr[i], arr[largest]);	
heapify(arr, n, largest)	
}	
}	